Software Requirement Document (SRD)

Campus Medical Appointment and Queue Management System

1. Introduction

1.1 Purpose

This document presents the functional and non functional requirements of the Campus Medical Appointment and Queue Management System.

The goal of the system is to replace the manual walk in procedure used at the university health center with a computerized platform that improves appointment scheduling queue handling and patient record management.

1.2 Scope

The proposed system will:

- Allow students faculty and staff to schedule medical appointments in advance.
- Provide digital queue tokens to patients who arrive without appointments.
- Support priority handling for emergency and follow up visits.
- Ensure doctors are not assigned more patients than available time slots.
- Maintain a history of patient visits.
- Send notifications regarding appointment confirmation and token progress.
- Allow doctors to view their daily schedule.
- Allow administrative staff to control doctor schedules.

The system will not:

- Offer online medical consultation.
- Replace the professional judgement of doctors.
- Connect with external hospitals or medical systems.

1.3 Definitions and Acronyms

Term	Meaning
Appointment	A planned medical visit assigned to a specific time slot
Token	A digital queue number given to walk in patients
Priority Case	Emergency or follow up case served before normal patients
Admin	Administrative staff responsible for schedule control
SRD	Software Requirement Document

2. Overall Description

2.1 Product Perspective

This application is a standalone solution created for the university health center. It replaces paper records and physical waiting queues.

The system centralizes appointment scheduling token distribution and patient record storage in one digital platform.

2.2 Product Functions

- Appointment scheduling module
- Digital token creation for walk in patients
- Priority queue control
- Doctor timetable management
- Patient visit history storage
- Role based user access
- Notification and alert system

2.3 User Classes

Patient Students faculty and staff who can book appointments check tokens and receive notifications.

Doctor Can review daily appointments and patient history.

Admin Manages doctor schedules appointments and records.

2.4 Constraints

- The system must operate within the university technical infrastructure.
- Doctors cannot be scheduled beyond their available time slots.
- Emergency and follow up patients must receive priority.
- System features must be accessible according to user roles.
- Walk in patients must be handled through digital tokens.

3. Requirements

3.1 Business Requirements

1. Reduce physical waiting queues at the health center.
2. Improve efficiency of appointment scheduling.
3. Maintain an organized and fair queue structure.
4. Store accurate patient visit information.
5. Support priority handling of emergency and follow up visits.

3.2 User Requirements

1. Patients can request appointments in advance.
2. Walk in patients receive a digital queue token.
3. Patients receive updates about appointment time and token progress.
4. Doctors can view their daily appointment schedule.
5. Administrators control doctor availability and scheduling.
6. Priority patients are served before normal queue patients.

3.3 Domain Requirements

1. Doctors operate using predefined time slots.
2. Each doctor can treat only one patient per slot.
3. Emergency and follow up visits are considered priority cases.
4. Patient records include visit history and assigned doctor details.

3.4 Functional Requirements

Appointment Handling

1. The system allows booking only for available time slots.
2. Scheduling conflicts for doctors must not occur.
3. Successful appointment booking must be confirmed.

Token Management

1. The system generates unique digital tokens for walk in patients.
2. The system maintains the correct queue order.
3. Token progress is updated dynamically.

Priority Handling

1. Emergency and follow up visits are marked as priority.
2. Priority patients are served before regular patients.

Role Based Usage

1. Authentication is provided according to user roles.
2. Patients access only their own information.
3. Doctors see only their assigned appointments.
4. Administrators manage schedules and records.

Record Management

1. Patient visit history is stored in the system.
2. Each visit contains assigned doctor information.

Notification

1. Patients receive confirmation messages for appointments.
2. Patients receive updates about queue progress.

3.5 Non Functional Requirements

1. System response time should remain within two seconds during normal use.
2. Patient information must remain secure and confidential.
3. The system operates during health center working hours.
4. Multiple users can access the system at the same time.
5. Data accuracy and consistency must be maintained.

4. System Models

4.1 Stakeholder List

Stakeholder	Interest
Students	Fast access to medical services
Faculty	Ability to book scheduled appointments
Staff	Reduced waiting time
Doctors	Organized daily schedule
Admin Staff	Simple schedule control
University Management	Efficient health center service

4.2 Use Case Diagram

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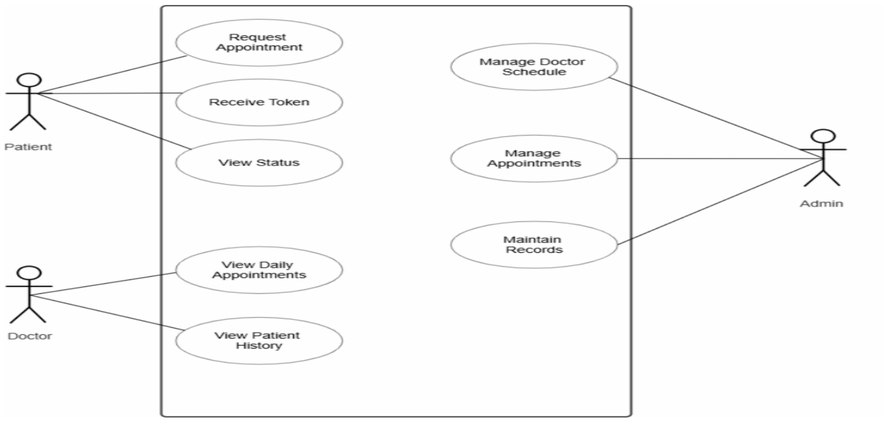

4.3 Swimlane Model

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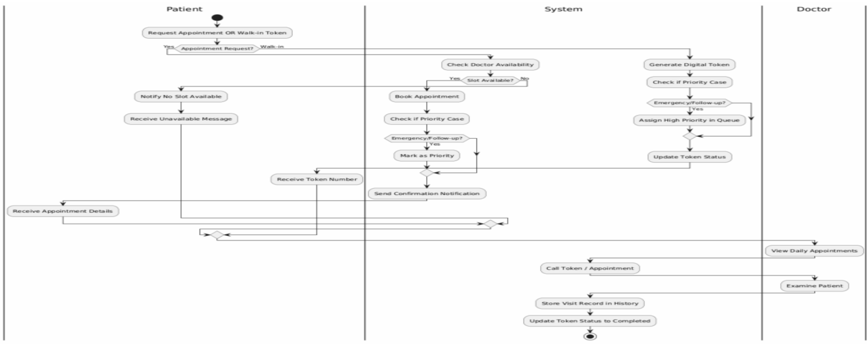

5. Requirement Traceability Matrix (RTM)

Requirement ID	Linked Use Case	Linked Actor
FR1	Request Appointment	Patient
FR4	Generate Token	Patient
FR7	Handle Priority Cases	System
FR9	Role Based Login	All Actors
FR13	Maintain Records	Admin
UR4	View Daily Appointments	Doctor